Algebraic Geometry

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This is a lecture note on Prof. Anna von Pippich's course *Algebraic Geometry* in summer semester, 2019. The course is mainly based on William Fulton's textbook *Algebraic Curves*.

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§0 Introduction

Algebraic geometry: study of systems of polynomial equations in several variables over a field. We consider only polynomials in 2 variables. Its set of solutions in k can be thought of as a curve. Goal of algebraic geometry is to study the properties of curves by just looking at polynomials.

Those curves may have different shapes: boundness, number of components, sigularity, connnectness...

Theorem (Fermat's last theorem)**.** *For $n \in \mathbb{N}, n > 2$, the curve $x^n + y^n - 1 = 0$ over $\mathbb{Q}$ has only trivial solutions (that is, $x = 0$ or $y = 0$).*

Goals of the class:

1. **local study**: let k be algebraic closed field, we want to know the intersection of two algebraic curves in the affine plane $\mathbb{A}^2(k)$.

   Let $F, G \in k[X, Y], V(F) = \{ x \in \mathbb{A}^2(k) \mid F(x) = 0 \}$ and similar for G. We want to understand $V(F) \cap V(G)$.

   If $V(F), V(G)$ has no common components then $\#V(F) \cap V(G) < \infty$

   (a) If $P \in V(F) \cap V(G)$ then $I(P, F \cap G)$ measures the intersections of P.

   (b) If $V(F), V(G)$ intersect transversally in p then $I(P, F \cap G) = 1$.
   Question: what if for some 'weird' cases?

2. **Global study**: work in $\mathbb{P}^2(k)$, we have

   Theorem (Bézout)**.** *$F, G \in k[X, Y]$ homogeneous, if $V(F), V(G)$ have no common component, then*

   $$\sum_{P \in V(F) \cap V(G)} I(P, F \cap G) = \deg(F) \cdot \deg(G)$$

   Later in the course we will deal with Riemann-Roch theorem.

§1 Basic Preliminaries

§1.1 Affine Algebraic Sets

Let k be a field, $\mathbb{A}^n(k) = \{ n\text{-tuples of elements of } k \}$, elements are called points.

Definition (affine zero locus)**.** Let $S \subset k[X_1, ..., X_n]$, $V(S) = \{ P \in \mathbb{A}^n \mid F(P) = 0, \forall F \in S \} = \cap_{F \in S} V(F)$ is called the affine zero locus of S.

If $S = \{ F_1, ..., F_n \}$, for simplicity we write $V(S) = V(F_1, ..., F_n)$.

Definition (affine algebraic set)**.** $X \subset \mathbb{A}^n(k)$ is an (affine) algebraic set, if $X = V(S)$ for some $S \subset k[X_1, ..., X_n]$

Remark.

1. Let $S \subset k[X_1, ..., X_n]$, $I = \langle S \rangle$ be the ideal generated by S, then $V(S) = V(I)$.
2. Consider all the algebraic sets of $\mathbb{A}^n(k)$ as closed subsets, this gives a topology on $\mathbb{A}^n(k)$, namely the **Zariski topology**.

   Note that we have $V(I \cap J) = V(I) \cup V(J)$ and $V(\cup_\alpha I_\alpha) = \cap_\alpha I_\alpha$
3. Any finite subset of $\mathbb{A}^n(k)$ is algebraic set.

Definition (ideal)**.** Let $V \subset \mathbb{A}^n(k)$, $I(V) = \{ F \in k[X_1, ..., X_n] \mid F(P) = 0, \forall P \in V \}$ is an ideal, called the ideal of V.

Remark.

1. $I(X)$ is an ideal of $k[X_1, ..., X_n]$.
2. $I(X)$ is radical.

Theorem (Hilbert basic theorem)**.** *Let R be Noetherian ring, then $R[X]$ is Noetherian.*

Corollary. *Every algebraic set is the intersection of finite number of hypersurfaces.*

Proof. $k[X_1, ..., X_n]$ is Noetherian, since any field and any PID is Noetherian. Therefore, Any ideal I is finitely generated. Let V be algebraic set, $V = V(S) = V(I)$ where $I = \langle S \rangle$, I finitely generated, so $I = \langle F_1, ..., F_n \rangle$, $V = \cap_{i=1}^n V(F_i)$. □

§1.2 Irreducible Components

Definition. An algebriac set $V \in \mathbb{A}^n(k)$ is **reducible** if there exists algebraic sets $V_1, V_2 \in \mathbb{A}^n(k)$ such that $V = V_1 \cup V_2$. Otherwise V is called **irreducible**.

Lemma. *An algebraic set $V \subset \mathbb{A}^n(k)$ is irreducible $\Leftrightarrow$ $I(V)$ is a prime ideal.*

Recall. Any (non-empty) collection of algebraic sets in $\mathbb{A}^n(k)$ has a minimal member.

Proof. Consider such a collection $\{ V_\alpha \}$. Since $k[X_1, ..., X_n]$ is Noetherian, $\{ I(V_\alpha) \}$ has a maximal member, which corresponds to a minimal member of $\{ V_\alpha \}$.

(maximal: in the sense not a subset of any other ideal; minimal: in the sense does not contain any other algebraic set). □

Proposition. *Let $V \subset \mathbb{A}^n(k)$ be an algebraic set, then it has a decomposition $V = V_1 \cup ... \cup V_n$ $(*)$ with:*

1. *$V_j \subset \mathbb{A}^n(k)$ irreducible, called **irreducible components** of V.*
2. *$V_j \not\subset V_i$, if $i \neq j$.*
3. *The decomposition is unique up to a change of order.*

Proof.
1. Let $S = \{$ algebraic sets $V \subset \mathbb{A}^n(k) \mid V$ has no decomposition of the form $(*)$ $\}$, then it has a minimal member V_0 that is not reducible. Then $V_0 = V_1 \cup V_2$. Since V_0 is minimal, $V_1, V_2 \notin S$, so $V_1 = V_{11} \cup ... \cup V_{1m_1}, V_2 = V_{21} \cup ... \cup V_{2m_2}$, then $V_0 = V_{11} \cup ... \cup V_{1m_1} \cup V_{1m_1} \cup V_{21} \cup ... \cup V_{2m_2}$ has the form $(*)$, this is a contradiction.
2. Simply throw away any V_i that is contained in some other V_j.
3. If there is another decomposition $V = W_1 \cup ... \cup W_m$, then $V_i = \cup_j (W_j \cap V_i)$, so $V_i \subset W_j$ for some j (since V_i irreducible). Similarly $W_j \subset V_k$ for some k, then $V_i \subset W_j \subset V_k \Rightarrow V_i = W_j = V_k$. □

§1.3 Algebraic Subsets of the Plane

Proposition. *Let $F, G \in K[X, Y]$ be polynomials with no common factors, then $V(F, G) = V(F) \cap V(G)$ is a finite set.*

Proof. Since F, G has no common factors in $k[X, Y]$, they have no common factors in $k(X)[Y]$. $k(X)[Y]$ is a PID, $(F, G) = (1)$ in $k(X)[Y]$, so there exists $R, S \in k(X)[Y]$ such that $RF + SG = 1$. There is a non-zero $D \in k[X]$ such that $DR = A, DS = B$ in $k[X, Y]$, therefore $AF + BG = D$. If $P = (a, b) \in V(F, G)$ then $D(a) = 0$. Since D has only finite zeros, we conclude $V(F, G)$ is finite. □

Corollary. *If $F \in k[X, Y]$ is irreducible and $V(F)$ is infinite, then $I(V(F)) = (F)$, and $V(F)$ is irreducible.*

Proof. $I(V(F)) \subset (F)$ is clear. If $G \in I(V(F))$, then $V(F, G)$ is infinite, so F is a factor of G, $G \in (F)$. Therefore, $I(V(F)) \supset (F)$. By lemma, $V(F)$ is irreducible since $I(V(F))$ is prime. □

Corollary. *Let k be infinite, then the irreducible algebraic subsets of $\mathbb{A}^2(k)$ are: ① $\mathbb{A}^2(k)$ ② ϕ ③ points ④ $V(F)$ for $F \in k[X, Y]$ irreducible and $V(F)$ infinite.*

Proof. Let V be an irreducible algebraic set. If V is finite, then V must be ② or ③. Now let V be infinite, consider $I(V)$. If $I(V) = \{0\}$, $V = \mathbb{A}^2(k)$ is ①; otherwise there is a non-constant polynomial $F \in I(V)$. V is irreducible means $I(V)$ is prime, we may assume F to be irreducible, otherwise we choose the irreducible factor of it, which is also in $I(V)$.

We claim that $I(V) = (F)$. $I(V) \supset (F)$ is clear. Suppose $I(V) \not\subset (F)$, there exists some $G \in I(V)$ such that F, G has no common factors, then $V(F, G)$ is finite. However, $V \subset V(F, G)$ is infinite!

$I(V) = (F)$, then we may choose $V = V(F)$ for V infinite, which is ④. □

Corollary. *Let k be algebraically closed, $F \in k[X,Y]$ is non-constant polynomial. Let $F = F_1^{n_1}...F_r^{n_r}$ be the decomposition of F into irreducible polynomials, then*

1. *$V(F) = V(F_1) \cap ... \cap V(F_r)$ is the decomposition of $V(F)$ into irreducible components.*
2. *$I(V(F)) = (F_1...F_r)$.*

Proof. TO BE ADDED □

§1.4 Hilbert's Nullstellensatz

Let k be an algebraically closed field.

Proposition (weak Nullstellensatz, algebraic version)**.** *Let A be a finitely generated k-algebra, $m \subset A$ be a maximal ideal, then $A/m \cong k$.*

Proof. Omitted. □

Proposition (weak Nullstellensatz, geometric version)**.** *Let $I \subset k[X_1, ..., X_n]$ be a proper ideal, then $V(I) \neq \phi$.*

Proof. Since I is proper ideal, there exists a maximal ideal m of $A = k[X_1, ..., X_n]$ with $I \subset m$. By algebraic version of weak Nullstellensatz, $A/m \cong k$. Let $\overline{X}_j \leftrightarrow x_j$ be the isomorphism. We let $P = (x_1, ..., x_n) \in V(I)$, and $m_P = (X_1 - x_1, ..., X_n - x_n) \subset m$. Since m_P is maximal, $m_P = m$, so $P \in V(m_P) = V(m) \subset I$. Then $V(I)$ has at least an element P. □

Corollary. *Let $I \subset k[X_1, ..., X_n]$ be a proper ideal, then there is a bijection*

$$\begin{array}{ccc} V(I) & \cong & \mathrm{Max}(k[X_1, ..., X_n]/I) \\ (x_1, ...x_n) & \leftrightarrow & (X_1 - x_1, ..., X_n - x_n) \mod I \end{array}$$

where $\mathrm{Max}(R)$ *denoted the maximal ideals of a ring R.*

Proof. Injectivity is clear.

Surjectivity: we have $k[X_1, ..., X_n]/I = k[\overline{X}_1, ..., \overline{X}_n]$ where $\overline{X}_i = X_i \mod I$. Let $\overline{m} \in \mathrm{Max}(k[X_1, ..., X_n])$, by algebraic version of weak Nullstellensatz we have a point $P = (x_1, ..., x_n) \in V(I)$ such that $\overline{m}_P = (X_1 - x_1, ..., X_n - x_n) \mod I \subset \overline{m}$. Then $\overline{m}_P = \overline{m}$ and $P \in V(\overline{m}_P) \subset V(I)$. □

Proposition (strong Nullstellensatz)**.** *Let $I \subset k[X_1, ..., X_n]$ be an ideal, then $I(V(I)) = r(I)$.*

Proof. It is clear that $r(I) \subset I(V(I))$. We show $r(I) \supset I(V(I))$.

Let $f \notin r(I)$. From sheet 1 we have a result:

$$\boxed{r(I) = \bigcap_{\mathfrak{p} \supset I, \mathfrak{p} \text{ is prime}} \mathfrak{p}}$$

that is, $r(I)$ is the intersection of all prime ideals that contain I as a subset. So $f \notin r(I)$ means that there exists a prime ideal $\mathfrak{p} \supset I$ such that $f \notin \mathfrak{p}$. Let $A = k[X_1, ..., X_n], B = A/\mathfrak{p}, C = B[\frac{1}{\overline{f}}]$, where $\overline{f}$ is the image of f in B. We have compositions of maps:

$$\begin{array}{ccccccc} A & \to & B & \to & C & \to & k(\cong C/m) \\ g & \mapsto & \overline{g} & \mapsto & \frac{\overline{g}}{1} & \mapsto & \overline{\overline{g}} \end{array}$$

Due to the preceeding corollary, there exists some maximal ideal m such that $C/m \cong k$, which is the last map.

We denote the image of $X_j \in A$ in k by x_j. We claim that $P = (x_1, ..., x_n)$ lies in $V(I)$: for all $h \in I \subset \mathfrak{p}$, we have $\overline{h(x_1, ..., x_n)} = h(\overline{X_1}, ..., \overline{X_n}) = \overline{0}$ in B, hence $P \in V(I)$.

Now we consider f. $f \notin \mathfrak{p}$, so $\overline{f(x_1, ..., x_n)} = \overline{f(X_1, ..., X_n)} \neq 0$ in B, so $f(P) \neq 0$, which means $f \notin I(V(I))$. We conclude that $r(I) \supset I(V(I))$. $\square$

Corollary. *We have bijections ($\rightarrow$ by V, $\leftarrow$ by I):*

radical ideals of $k[X_1, ..., X_n]$	$\longleftrightarrow$	*algebraic sets of* $\mathbb{A}^n(k)$
prime ideals of $k[X_1, ..., X_n]$	$\longleftrightarrow$	*irreducible algebraic sets of* $\mathbb{A}^n(k)$
maximal ideals of $k[X_1, ..., X_n]$	$\longleftrightarrow$	*points of* $\mathbb{A}^n(k)$

§2 Affine Varieties

§2.1 The Coordinate Ring

Definition. An **affine variety** is an irreducible algebraic set.

Definition. If $V \subset \mathbb{A}^n(k), V \neq \phi$ is an affine variety, then $I(V)$ is a prime ideal. Hence $\mathcal{R}(V) := [X_1, ..., X_n]/I(V)$ is an integral domain, called the **coorinate ring** of V.

Let $V \subset \mathbb{A}^n(k)$ be an affine variety, we consider $\mathcal{F}(V,k) := \{ f \colon V \to k \}$ with the natural ring structure. We identify k with the subring of $\mathcal{F}(V,k)$ consisting of constant functions.

Definition. A function $f \in \mathcal{F}(V,k)$ is called a **polynomial function** if there is a polynomial $F \in k[X_1, ..., X_n]$ such that $f(x_1, ..., x_n) = F(x_1, ..., x_n)$ for all $(x_1, ..., x_n) \in V$.

Remark. $F, G \in k[X_1, ..., X_n]$ determine the same polynomial function in $\mathcal{F}(V,k) \quad \Leftrightarrow \quad F - G \in I(V)$. Hence $\mathcal{R}(V)$ can be identified with subring of $\mathcal{F}(V,k)$ consisting of all polynomial functions.

§2.2 Polynomial Maps

In this section we let $V \in \mathbb{A}^n(k), W \in \mathbb{A}^m(k)$ be affine varieties.

Definition. A map $\phi \colon V \to W$ is a polynomial map if there are polynomials $T_1, ..., T_m \in k[X_1, ..., X_n]$ such that $\phi(x_1, ..., x_n) = (T_1(x_1, ..., x_n), ..., T_m(x_1, ..., x_n))$ for all $(x_1, ..., x_n) \in V$.

Remark. Polynomial maps are continuous in the Zariski topology.

Example (affine changes of coordinates). $\phi \colon \mathbb{A}^n(k) \to \mathbb{A}^m(k), (a_i) \mapsto (\sum \alpha_{ij} a_j + \beta_i)$ is a polynomial map.

Remark. A polynomial map $\phi \colon V \to W$ induces a map $\phi^* \colon \mathcal{F}(W,k) \to \mathcal{F}(V,k)$. Note that $\phi^*(\mathcal{R}(W)) \subset \mathcal{R}(V)$, we have a contravariant functor $\mathcal{R}$ from category of affine varieties to category of finitely generated k-algebras that are integral domains.

Proposition. *There is a 1-1 correspondence between $\phi \colon V \to W$ and $\phi^* \colon \mathcal{R}(W) \to \mathcal{R}(V)$, and any such ϕ is the restriction of a polynomial map from $\mathbb{A}^n(k) \to \mathbb{A}^m(k)$. That is,*

$$\begin{tikzcd} V \arrow[r, "\phi"] \arrow[d, hook] & W \arrow[d, hook] \\ \mathbb{A}^n(k) \arrow[r, dashed, "T"] & \mathbb{A}^m(k) \end{tikzcd} \qquad \begin{tikzcd} \mathcal{R}(W) \arrow[r, "\phi^*"] & \mathcal{R}(V) \\ \mathcal{R}(\mathbb{A}^m(k)) \arrow[u, two heads] \arrow[r, dashed, "T^*"] & \mathcal{R}(\mathbb{A}^m(k)) \arrow[u, two heads] \end{tikzcd}$$

where $\phi = T|_V$. Note that $\mathcal{R}(\mathbb{A}^m(k)) = k[X_1, ..., X_m]$.

Proof. TO BE ADDED □

Definition. A polynomial map $\phi \colon V \to W$ is an isomorphism if there is a polynomial map $\psi \colon W \to V$ such that $\phi \circ \psi = \mathrm{id}_W, \psi \circ \phi = \mathrm{id}_V$. I write $V \cong W$ in this case.

Remark. $V \cong W \quad \Leftrightarrow \quad \mathcal{R}(V) \cong \mathcal{R}(W)$

§2.3 Rational Functions and Local Rings

In this section we let $V \subset \mathbb{A}^n(k)$ be an affine variety and $P \in V$ be a point of V.

Definition. The quotient field of $\mathcal{R}(V)$ is called the **field of rational functions on V**, denoted by $\mathcal{K}(V)$. Elements of $\mathcal{K}(V)$ are called **rational functions on V**.

Definition. We say $f \in \mathcal{K}(V)$ is defined at $P \in V$, if there are $a, b \in \mathcal{R}(V)$ such that $f = \frac{a}{b}$ and $b(P) \neq 0$.

Remark. In general there may be different choices of a, b such that $f = \frac{a}{b}$. However, if $\mathcal{R}(V)$ is UFD, then there is a unique pair (a, b) up to units such that a, b has no common factors. In this case f is defined at P if and only if $b(P) \neq 0$.

Definition. $\mathcal{O}_P(V) := \{ f \in \mathcal{K}(V) \mid f \text{ is defined at } P \}$ is called the **local ring of V at P**.

Remark. We have following inclusion of *rings*:

$$k \subset \mathcal{R}(V) \subset \mathcal{O}_P(V) \subset \mathcal{K}(V)$$

Definition. Let $f \in \mathcal{O}_P(V)$. The **value of f at P** is defined as $f(P) = \frac{a(P)}{b(P)}$ where $f = \frac{a}{b}$ for $a, b \in \mathcal{R}(V), b \neq 0$. Check that this is independent of the representation of a, b.

Definition. $\mathcal{M}_P(V) := \{ f \in \mathcal{O}_P(V) \mid f(P) = 0 \}$ is called the **maximal ideal of V at P**.

Proposition. *$\mathcal{M}_P(V)$ is the unique maximal ideal in $\mathcal{O}_P(V)$.*

Proof. $\mathcal{O}_P(V) \to k, f \mapsto f(P)$ is surjective with kernal $\mathcal{M}_P(V)$. So $\mathcal{O}_P(V)/\mathcal{M}_P(V) \cong k$.

Let $f \in \mathcal{O}_P(V) - \mathcal{M}_P(V)$, then $f(P) \neq 0$, so there is $a, b \in \mathcal{R}(V), a \neq 0, b \neq 0$ such that $f = \frac{a}{b}$. Then f is a unit with inverse element $\frac{b}{a}$. □

Definition. A ring R is called a **local ring**, if R has a unique maximal ideal.

Remark. R local $\Leftrightarrow$ The set of non-units in R form an ideal, which is maximal.

Proposition. *$\mathcal{O}_P(V)$ is a Noetherian local ring.*

Proof. It remains to prove $\mathcal{O}_P(V)$ is Noetherian, i.e. every ideal I of $\mathcal{O}_P(V)$ is finitely generated. Since $\mathcal{R}(V)$ is Noetherian, we have $I \cap \mathcal{R}(V)$ is finitely generated, say, by $f_1, ..., f_r \in \mathcal{R}(V)$. We claim that $f_1, ..., f_r$ generate I over $\mathcal{O}_P(V)$.

Let $f \in I \subset \mathcal{O}_P(V)$, then there exist $a, b \in \mathcal{R}(V), b(P) \neq 0$ such that $f = \frac{a}{b}$. Then $bf = a \in \mathcal{R}(V)$. Since $b \in \mathcal{R}(V) \subset \mathcal{O}_P(V), f \in I$, we have $bf \in I$, then $bf \in I \cap \mathcal{R}(V)$. Then there exists $a_1, ..., a_r \in \mathcal{R}(V)$ such that $bf = \sum_i^r a_i f_i$, $f = \sum_{i=1}^r (\frac{a_i}{b}) f_i$, and $\frac{a_i}{b} \in \mathcal{O}_P(V)$. □

§2.4 Discrete Valuation Rings

Lemma. *Let R be a ring that is not a field, then the followings are equivalent:*

1. *R is Noetherian local ring and the maximal ideal M of R is principal.*
2. *There is a irreducible element $t \in R$ such that every non-zero $z \in R$ can be uniquely written in the form $z = u \cdot t^n$, for $u \in R^\times$ (units of R), and $n \in \mathbb{N}_{\geq 0}$.*

Proof.

- $(1) \Rightarrow (2)$: Let $M = (t)$ for $t \in R$ irreducible. We prove that any non-zero $z \in R$ can be written in the form given in (2).

  If $z \in R^\times$, then $z = z \times t^0$. Now we let $z \notin R^\times$, then $z \in M$, that is, there exists $z_1 \in R, z_1 \neq 0$ with $z = z_1 t$. If $z_1 \in R^\times$, we are done. Otherwise we find $z_2 \in R^\times, z_2 \neq 0$ such that $z_1 = z_2 t$. Then we have

  $$z = z_1 t = z_2 t^2 = ... = z_n t^n = ...$$

  which yields a chain of ideals

  $$(z_1) \subset (z_2) \subset ... \subset (z_n) \subset ...$$

  Since R is Noetherian, the sequence stops after finite steps, say n, then z_n is unit.

  Uniqueness: Let $z = ut^n = vt^m$ with $u, v \in R^\times$ and $n, m \in \mathbb{N}_{\geq 0}$. Without loss of generality we let $n \leq m$, then $ut^{n-m} = v \in R^\times$, $t^{n-m} \in R^\times, n - m = 0$.

- $(2) \Rightarrow (1)$: We show that the ideal $M = (t)$ is the unique maximal ideal. If $I \subset R$ is an ideal, then $I \subset M$ or I contains a unit. Further we may see that every ideal of R has the form $I = (t^n)$ for some $n \in \mathbb{N}_{\geq 0}$.

□

Definition. An integral domain satisfying the preceeding properties is called a **discrete valuation ring**, or **DVR**. The element t is called **uniformizing parameter**, which is unique up to a multiplication of unit.

Remark. Let R be DVR with uniformizing parameter t, K be the quotient field of R, M be *the* maximal ideal of R, then

$$\begin{aligned} K &= \{\, ut^n \mid u \in R^\times, n \in \mathbb{N} \,\} \\ R &= \{\, ut^n \mid u \in R^\times, n \in \mathbb{N}_{\geq 0} \,\} \\ M &= \{\, ut^n \mid u \in R^\times, n \in \mathbb{N}_{>0} \,\} \end{aligned}$$

§3 Affine Plane Curves

§3.1 Some Notations

Definition. Two polynomials $F, G \in k[X, Y]$ are equivalent if $F = \lambda G$ for some $\lambda \in k^\times$.

Definition. An **affine plain curve** is the zero locus of an equivalent class of a non-constant polynomial of $k[X, Y]$, or simply saying, the zero locus of a non-constant polynomial.

Definition. The **degree** of an affine plane curve is the degree of its defining polynomial.

Definition. If $F \in k[X, Y]$, F is not constant, then it has a decomposition $F = \prod_{j=1}^{r} F_j^{e_j}$ in irreducible factors F_j. Then we say F_j's the **(irreducible) components** of F, and e_j's the **multiplications** of F.

Definition. A component F_j is called **simple** if $e_j = 1$; otherwise it is called **multiple**.

If $F \in k[X, Y]$ is irreducible, then $V(F)$ is a variety in $\mathbb{A}^n(k)$. For simplicity we write $\mathcal{R}(F), \mathcal{K}(F), \mathcal{O}_P(F)$ for $\mathcal{R}(V(F)), \mathcal{K}(V(F), \mathcal{O}_P(V(F)))$.

§3.2 Simple and Regular Points

Let $F \in k[X, Y], P = (a, b) \in V(F)$.

Definition. P is called a **simple** (or **regular**, **smooth**) **point**, if $(\frac{\partial F}{\partial X}, \frac{\partial F}{\partial Y})\big|_P \neq (0, 0)$. Otherwise P is called **singular** (or **multiple**) **point**.

Remark. If P is simple, then the tangent line at $V(F)$ in P is given by

$$\boxed{\left.\frac{\partial F}{\partial X}\right|_P (X - a) + \left.\frac{\partial F}{\partial Y}\right|_P (Y - b) = 0}$$

Definition. An affine plane curve is called **non-singular** if there is no singular point on the curve.

Remark. Let $F \in k[X, Y]$ be of degree n, then one can decompose F into homogeneous of degree j, called forms of degree j. If $F = F_m + ... + F_n$ is the decomposition, $m < m + 1 < ... < n$, let $Q = (0, 0)$, then m is called the **multiplicity of $V(F)$ at Q**, denoted by $m = m_Q(F)$. We have:

1. $Q \in V(F) \quad \Leftrightarrow \quad m_Q(F) > 0$
2. $Q \in V(F)$ simple $\quad \Leftrightarrow \quad m_Q(F) = 1$, in this case F_1 is the tangent line of $V(F)$ at Q.
3. $Q \in V(F)$ is called double (resp. triple, ...) if $m_Q(F) = 2$ (resp. 3, ...)

§A Appendix: Algebraic Preliminaries

§A.1 Ring

We have the following chain of class inclusion:

Commutative rings $\supset$ integral domains $\supset$ Integrally closed domain $\supset$ GCD domains $\supset$ Unique factorization domains (UFD) $\supset$ principal ideal domains (PID) $\supset$ Euclid domains $\supset$ fields.

§A.1.1 Integral Domain

An integral domain can be described by:

- A nonzero commutative ring with no nonzero zero devisor, which means nonzero elements are cancellable.
- A nonzero commutative ring where zero ideal $\{ 0 \}$ is prime.
- A subring of a field.

Some properties:

- A commmutative ring R is an integral domain $\Leftrightarrow$ $\{ 0 \}$ is prime.
- If R is a commutative ring and P is an ideal of R, then R/P is integral domain $\Leftrightarrow$ P is prime.
- If R is an integral domain, then so is $R[x]$.

§A.1.2 Integrally Closed Domain

Definition (field of fraction)**.** The **field of fraction** of an integral domain R is the smallest field in which it can be embedded in. Its elements are equivalence classes of $\frac{a}{b}, a, b \in R, b \neq 0$, with equivalent relation $\frac{a}{b} \sim \frac{c}{d} \Leftrightarrow ad = bc$.

Definition (integral element)**.** Let R be a commutative ring, $A \subset R$ a subring. $x \in R$ is **integral over** A, if x is a root of a monic polynomial over A, that is, there exists $n \geq 1$ and $a_{1,\ldots,n} \in A$ such that

$$x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0 = 0$$

If every $x \in R$ is integral over A, then one says R is integral over A, or R is an **integral extension** of A.

Definition (integral closure)**.** The **integral closure** of A in R (see the preceeding definition) is $\{ x \in R \mid x \text{ is integral over } A \}$. It is a subring of R that contains A!

Definition (integrally closed domain)**.** An **integrally closed domain** R is an integral domain whose integral closure in its field of fractions is R itself.